

Submission STAT1-PS2-SUB08

Question STAT1-PS2-Q01

scan faint / margin cut off

visible work:

$$\bar{x} = 33/5 = 6.60 \dots ?$$

last line is hard to read; maybe mean=6.60; sample variance=16.30

Question STAT1-PS2-Q02

scan faint / margin cut off

visible work:

$$P(D_1 \cap D_2) = (5/13)((5-1)/(13-1)) \dots ?$$

last line is hard to read; maybe 0.1282

Question STAT1-PS2-Q03

scan faint / margin cut off

visible work:

$$SE = \sqrt{(p(1-p)/n)} = \sqrt{(0.55(1-0.55)/40)} = 0.0787 \dots ?$$

last line is hard to read; maybe (0.396, 0.704); repeated-sampling interpretation

Question STAT1-PS2-Q04

scan faint / margin cut off

visible work:

Increase sample size n to reduce standard error.... ?

last line is hard to read; maybe increase n; lower confidence level or variability; note the coverage trade-off

Question STAT1-PS2-Q05

scan faint / margin cut off

visible work:

$0.03 < 0.05$, so reject H_0 under the stated rule.... ?

last line is hard to read; maybe reject H_0 ; evidence against H_0 , not proof that H_0 is false

Question STAT1-PS2-Q06

scan faint / margin cut off

visible work:

$$\bar{x} = 29/5 = 5.80 \dots ?$$

last line is hard to read; maybe mean=5.80; sample variance=16.70

Question STAT1-PS2-Q07

scan faint / margin cut off

visible work:

$$P(D_1 \cap D_2) = (5/19)((5-1)/(19-1)) \dots ?$$

last line is hard to read; maybe 0.0585

Question STAT1-PS2-Q08

scan faint / margin cut off

visible work:

$$SE = \sqrt{(p^*(1-p^*)/n)} = \sqrt{(0.45(1-0.45)/80)} = 0.0556... ?$$

last line is hard to read; maybe (0.341, 0.559); repeated-sampling interpretation